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Design

Note: the folder 6.3.0 contains the only external library used to implement this project, the GNU Multiple Precision Arithmetic Library).

Running the Project:

After running the executable (typing “./myprogram” in the folder “applied_crypto_hw_0” after make), the program will output the message:

```
Welcome! Please enter a sequence of newline separated commands, starting with setup_params!  
Here is an example sequence:  
    setup_params p g q  
    rsa_encrypt N publicKey message  
    rsa_decrypt N secretKey ciphertext  
    exit  
Please type exit to finish your command sequence.
```

Here is an example of a sequence of commands. Each command contains the command name and parameters separated by spaces (e.g. setup_params p g q), and commands themselves are separated by newline characters. To end the sequence, type exit:

```
Welcome! Please enter a sequence of newline separated commands, starting with setup_params!  
Here is an example sequence:  
    setup_params p g q  
    rsa_encrypt N publicKey message  
    rsa_decrypt N secretKey ciphertext  
    exit  
Please type exit to finish your command sequence.  
setup_params 0xFFFFFFFFFFFFFFFFC90FDAA22168C234C4C6628B80DC1CD129024E088A67CC74020BBEA63B139B225  
14A08798E3404DDEF9519B3CD3A431B302B0A6DF25F14374FE1356D6D51C245E485B576625E7EC6F44C42E9A63A3620F  
FFFFFFFFFFFFFFFF 0x2 0x7FFFFFFFFFFFFFFFFFE487ED5110B4611A62633145C06E0E68948127044533E63A0105DF531D  
89CD9128A5043CC71A026EF7CA8CD9E69D218D98158536F92F8A1BA7F09AB6B6A8E122F242DABB312F3F637A262174D3  
1D1B107FFFFFFFFFFFFFFFF  
dh_key_exchange 0x11d292ac4a2966be615af2da42f8d84fcb26f47e71b520fbb6d825b9c4b8534d90179a778ad6a7  
2e1670b59b23e88a296c7040c47c86ab8abe953669ffbc587d5a43d698c1abc3312dbc53767594d75192c6ce321339d  
e2400f8ff92f577a3  
exit
```

Results:

Here is an example of what the output would look like (the output for the prior sequence of commands). Each result (e.g. Diffie Hellman Key Exchange result) will be separated from other results by newline. The output will also have labels, and if there are multiple outputs, they will be separated by newline. :

```
Welcome! Please enter a sequence of newline separated commands, starting with setup_params!
Here is an example sequence:
    setup_params p g q
    rsa_encrypt N publicKey message
    rsa_decrypt N secretKey ciphertext
    exit
Please type exit to finish your command sequence.
setup_params 0xFFFFFFFFFFFFFFFFC90FDAA22168C234C4C6628B80DC1CD129024E088A67CC74020BBEA63B139B225
14A08798E3404DDEF9519B3CD3A431B302B0A6DF25F14374FE1356D6D51C245E485B576625E7EC6F44C42E9A63A3620F
FFFFFFFFFFFFFFFF 0x2 0x7FFFFFFFFFFFFFFFFFE487ED5110B4611A62633145C06E0E68948127044533E63A0105DF531D
89CD9128A5043CC71A026EF7CA8CD9E69D218D98158536F92F8A1BA7F09AB6B6A8E122F242DABB312F3F637A262174D3
1D1B107FFFFFFFFFFFFFFFF
dh_key_exchange 0x11d292ac4a2966be615af2da42f8d84fcb26f47e71b520fbb6d825b9c4b8534d90179a778ad6a7
2e1670b59b23e88a296c7040c47c86ab8abe953669ffbc587d5a43d698c1abc3312dbc53767594d75192c6ce321339d
e2400f8ff92f577a3
exit

Results:

Diffie-Hellman Key Exchange:
Private key kb: 131cc1de83cd19e23f360518d758bac6c636851ba9901177fcaad4913505931f395556d8cc57e3ac
152779840c522fc4e72c5079f8a287ea42819d34eff27fd967410429a0e0c2a8a47284d9ed56ce35ffc9985121b2e323
c1e8a4e6d8d0dd5d
Shared secret key: 155106ab9014d7de31375c5e59eaa35d8147a883badd363785412446170b594982de32874ab83
e095c949ab75bf2365e9897573a115a852c9e272507ca5eb4dcfe3cacb346a04455762ab80d192c16b1866140ed1341a
093a7fcf7d450724523
```